

Card (1995) Application: Diagnostic Findings

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Outcome weights, ESS diagnostics, and the MLE vs. CBPS propensity score comparison

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Context. This note summarises the empirical diagnostics run on the Card (1995) dataset as part of the thesis analysis in Chapter 5. The goal was to understand *why* the kappa-based LATE estimators produce near-zero effective sample sizes (ESS) in this dataset, and whether the problem can be attributed to (a) a structural overlap failure in the data, (b) the choice of propensity score estimator (MLE logit vs. CBPS), or (c) the choice of covariate specification (Card vs. Kitagawa).

Setup. $Z = \text{nearc4}$ (proximity to four-year college), $D_1 = \text{somecol}$ ($\text{educ} > 12$), $D_2 = \text{educ16}$ ($\text{educ} \geq 16$), $Y = \log(\text{wage})$, $N = 3,010$. Two covariate specs: *Card* (full controls incl. 8 region dummies) and *Kitagawa* (parsimonious: black, south, smsa, smsa66, south66). Two propensity score estimators: logit MLE and CBPS (custom Newton solver from `functions_card.R`).

Root Cause Analysis: Why ESS Collapses

Three compounding causes were identified, each quantified from the descriptive diagnostics.

Cause 1 — Weak first stage

Treatment	Adjusted first stage	First-stage F
D_1 (somecol)	0.064	39.3
D_2 (educ16)	0.030	15.6

The overall adjusted first stage is small. A small $\hat{\pi}_c$ means the instrument explains little variation in treatment take-up; kappa weights scale inversely with $\hat{\pi}_c$, so all identification weight is forced onto the few complier-like observations.

Cause 2 — Tiny complier share

Treatment	$\hat{\pi}_c$	Always-taker share	Never-taker share
D_1 (somecol)	12.2%	17.1%	70.7%
D_2 (educ16)	6.9%	—	93.1%

By Abadie (2003), Lemma 2.1, always-takers and never-takers receive negative κ_i weights by construction. With 87–93% non-compliers, the negative weight share is expected to be around 31–33%, which is exactly what is observed.

Cause 3 — Skewed $P(Z = 1 | X)$ distribution

The Card specification produces a highly right-skewed instrument propensity score:

Spec / Estimator	Min	Median	Max	% above 0.93
Card MLE	0.190	0.791	0.949	12.4%
Card CBPS	0.193	0.776	0.957	4.4%
Kitagawa MLE	0.253	0.798	0.907	0.0%
Kitagawa CBPS	0.251	0.802	0.913	0.0%

The kappa weight for observation i contains the term $D_i(1 - Z_i)/(1 - p_i)$, which diverges as $p_i \rightarrow 1$. When $p_i = 0.933$ and $(Z_i = 0, D_i = 1)$, this gives $\kappa_i = 1 - 1/(1 - 0.933) \approx -13.9$.

Additional issue — First-stage heterogeneity

The adjusted first stage varies dramatically across regions and subgroups:

Subgroup / Region	Adj. first stage (D_1)	Ratio to overall
Non-black	0.078	+1.2×
Black	0.018	+0.3×
Urban (SMSA)	0.091	+1.4×
Rural	0.018	+0.3×
reg668 (Mountain)	-0.213	-3.3×
reg666 (Pacific)	0.018	+0.3×
reg661 (New England)	-0.053	-0.8×

In regions 661 and 668, the adjusted first stage is negative — the instrument runs backwards. Including these observations actively distorts the kappa weight distribution.

ESS and Outcome Weight Diagnostics

Table 1 reports the outcome weight diagnostics for $\hat{\tau}_u$ and $\hat{\tau}_{a,10}$ across all four specification-estimator combinations.

Table 1: Outcome weight diagnostics: ESS, % negative weights, $\max|\omega_i|$

Estimator	$\sum \omega_i$	ESS	% neg.	$\max \omega_i $
<i>Treatment $D_1 = somecol$</i>				
$\hat{\tau}_u$ Card MLE	0	5	31.8	0.049
$\hat{\tau}_{a,10}$ Card MLE	0	5	31.8	0.047
$\hat{\tau}_u$ Card CBPS	0	5	31.8	0.068
$\hat{\tau}_{a,10}$ Card CBPS	0	5	31.8	0.068
$\hat{\tau}_u$ Kit MLE	0	3	31.8	0.044
$\hat{\tau}_{a,10}$ Kit MLE	0	3	31.8	0.053
$\hat{\tau}_u$ Kit CBPS	0	3	31.8	0.046
$\hat{\tau}_{a,10}$ Kit CBPS	0	3	31.8	0.046
<i>Treatment $D_2 = educ16$</i>				
$\hat{\tau}_u$ Card MLE	0	1	31.8	0.089
$\hat{\tau}_{a,10}$ Card MLE	0	2	31.8	0.087
$\hat{\tau}_u$ Card CBPS	0	1	31.8	0.129
$\hat{\tau}_{a,10}$ Card CBPS	0	1	31.8	0.129
$\hat{\tau}_u$ Kit MLE	0	1	31.8	0.078
$\hat{\tau}_{a,10}$ Kit MLE	0	0	31.8	0.134
$\hat{\tau}_u$ Kit CBPS	0	1	31.8	0.081
$\hat{\tau}_{a,10}$ Kit CBPS	0	1	31.8	0.081

Key observation. $\sum \omega_i = 0$ holds exactly for all translation-invariant estimators, confirming correct implementation. ESS ranges from 1 to 5 out of $N = 3,010$. *ESS is identical across MLE and CBPS within each spec*, demonstrating that the weight collapse is structural.

MLE vs. CBPS: What Changes and What Does Not

The extreme-weight mechanism

The most negative kappa observations all share the same profile: $p_i \approx 0.933$, $Z_i = 0$, $D_i = 1$. These are individuals in region 662 (Northeast), where 431 of 484 observations have $Z = 1$. The logit model correctly predicts $p \approx 0.93$ for all in this cell; nine of them happen to have $Z = 0$ and $D = 1$, yielding:

$$\kappa_i = 1 - \frac{D_i(1 - Z_i)}{1 - p_i} = 1 - \frac{1}{1 - 0.933} \approx -13.9$$

The top 5 such observations account for 2.3% of the total $|\kappa|$ mass.

What CBPS changes

Quantity	Card MLE	Card CBPS
% observations with $p > 0.93$	12.4%	4.4%
$\min(\kappa)$	-13.87	-11.56
% $\kappa < -5$	1.83%	1.53%
% $\kappa < -10$	0.37%	0.37%
ESS (D_1 somecol, $\hat{\tau}_u$)	5	5

CBPS reduces the proportion of observations above the critical threshold from 12.4% to 4.4% by enforcing covariate balance in the instrument propensity score. However, the share of $\kappa < -10$ is identical under both estimators (0.37%), and the ESS is unchanged. The same 11 most extreme observations dominate the weight distribution regardless of estimation method.

What CBPS does not change

CBPS cannot create compliers that are not present in the data. The ESS formula, $ESS = 1 / \sum_i \omega_i^2$, collapses because the distribution of ω_i has a fat-tailed structure driven by the tiny complier share ($\hat{\pi}_c = 12.2\%$) and the near-degenerate instrument propensity in region 662. This is a property of the data generating process, not of the propensity score estimator.

Specification vs. estimator: effect on point estimates

Table 2 shows $\hat{\tau}_u$ across all combinations.

Table 2: $\hat{\tau}_u$ point estimates across specifications and estimators

Treatment	Card MLE	Card CBPS	Kit MLE	Kit CBPS
D_1 (somecol)	0.290	0.249	0.356	0.331
D_2 (educ16)	0.529	0.471	0.628	0.589
Within-spec spread (MLE vs CBPS)	0.041 (D_1) / 0.058 (D_2)		0.025 (D_1) / 0.039 (D_2)	
Across-spec spread (Card vs Kit)	0.066 (D_1) / 0.099 (D_2)			

The covariate specification (Card vs. Kitagawa) has roughly twice the impact on point estimates as the propensity score estimator (MLE vs. CBPS). The larger estimates under the Kitagawa specification likely reflect that removing region dummies changes the conditioning set and therefore the implicit complier population being targeted.

Thesis Implications

For RQ2 (empirical comparison DML vs. kappa)

The DML Wald-AIPW estimators (PLR-IV, Wald-AIPW) from the Card analysis also produce $ESS \approx 2-3$, consistent with the kappa estimators. This suggests the weight collapse is not an artefact of the kappa weighting approach but a genuine feature of this identification design. The PIVE/kappa framework simply makes it visible.

For RQ3 (practitioner guidance)

The Card application provides a clear case study for the practitioner guidance chapter:

1. **ESS is invariant to propensity score estimator.** Switching from MLE to CBPS does not improve ESS. If ESS is near zero, the problem is in the data (complier share, instrument heterogeneity), not in the model.
2. **Covariate specification dominates.** The Card spec's 8 region dummies create near-degenerate propensity cells that amplify kappa weights. Region-specific first-stage diagnostics (e.g. adj. FS = -0.213 for reg668) should be reported alongside ESS as part of the standard diagnostic suite.
3. **The kappa framework surfaces what 2SLS conceals.** A 2SLS with $F = 39$ looks acceptable in the first-stage table; the ESS diagnostic reveals the underlying identification rests on ≈ 5 effective observations. This is the central argument for the PIVE/kappa pipeline.
4. **Recommended specification for the Card analysis:** somecol treatment (D_1), Kitagawa covariate spec (no region dummies), CBPS propensity score, with regions 661 and 668 flagged or excluded as robustness checks. educ16 (D_2) should be reported as a secondary case to illustrate the ESS collapse under an even weaker instrument ($F = 15.6$, $\hat{\pi}_c = 6.9\%$).

Open questions for supervision

1. Should the Card application be retained as a full chapter, or reframed as a *pathological case* that demonstrates the limits of the instrument rather than a clean replication of Card's findings?
2. How should the Block D complier covariate profiles be presented? With the marginal propensity score the estimates are more stable (min $\kappa = -2.15$, 4 flags) than with the Card logit (min $\kappa = -14.6$, 7-8 flags). The instability itself is a finding.
3. Is the ESS = 1-5 result severe enough to warrant a recommendation that practitioners *not* use this instrument for LATE estimation, and if so, how should this be framed relative to Card's original 2SLS results?

Summary Table

Table 3: Summary: three causes, three non-remedies, one structural conclusion

Cause	Magnitude	Attempted remedy	Result
Weak first stage	FS = 0.064 / 0.030	Focus on D_1 only	FS = 0.064, $F = 39$ — borderline
Tiny complier share	$\hat{\pi}_c = 12\% / 7\%$	Urban / non-black subsample	FS rises to 0.091 in urban
Skewed $P(Z = 1 X)$	63.9% have $p > 0.70$	CBPS instead of MLE	ESS unchanged; structural
First-stage heterogeneity	reg668: FS = -0.213	Drop reg661, reg668	Reduces extreme κ cells

Structural conclusion. The near-zero ESS in the Card application is a property of the instrument (`nearc4`) in this sample, not of the estimator or covariate specification. The kappa/PIVE framework does not *cause* the problem — it *diagnoses* it. A standard 2SLS with the same data would report a point estimate without any indication that the identification rests on fewer than 5 effective observations.